UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI

ICT LAB

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A HYBRID TEMPLATE-BASED AND DEEP-LEARNING FRAMEWORK FOR RNA THREE-DIMENSIONAL STRUCTURE PREDICTION WITH TEMPORAL LEAKAGE CONTROL AND GEOMETRY REFINEMENT

MASTER THESIS

DATA SCIENCE

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HANOI, 2026

Declaration

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Acknowledgements

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Abstract

Predicting the three-dimensional structure of RNA from sequence remains difficult because the conformational space is large, experimental structures are limited, and RNA is intrinsically flexible [1, 11, 10]. This thesis develops a hybrid pipeline that combines time-safe template-based modeling, DRfold2 as an independent candidate source, and geometry refinement for local correction. The system produces three template-derived structures and two DRfold2 structures, then refines each candidate without access to native labels. This design matches the best-of-five evaluation used in the Stanford RNA 3D Folding competition [12, 27].

The evidence is separated into local component studies and external hidden-set evaluation. The local protocol uses 60 temporally earlier RNAs to estimate geometric priors, 20 RNAs to select the configuration, and 20 family-disjoint newer RNAs for a single confirmatory evaluation. In the TBM component study, MMseqs2 returned a template for 8 of 20 calibration targets, whereas composite search produced candidates for all 20. The composite score therefore increased retrieval coverage, although its full weighted form did not clearly outperform global alignment alone.

On the 20 held-out RNAs, the best available score from three TBM candidates averaged 0.565183, and the corresponding score from two DRfold2 candidates averaged 0.502627. Their union reached 0.588304, a gain of 0.023121 with a target-bootstrap 95% confidence interval of $[0.006444, 0.047519]$. DRfold2 supplied the better source on 8 of 20 targets, indicating complementarity rather than higher average accuracy. Geometry refinement increased C1′-lDDT by 0.006660 and reduced SW-RMSD9 by 0.040386 Å, while TM-score changed by -0.001107 with a confidence interval that included zero. The refinement therefore corrected local relationships while approximately preserving the global fold.

On Kaggle, the time-safe TBM pipeline scored 0.60084 on the public set and 0.60175 on the private set. The final three-TBM, two-DRfold2 pipeline with geometry refinement reached 0.62809 and 0.61390, respectively, exceeding the reported 0.59298 private score of the reference TBM-only notebook. Because the final submission changed both candidate composition and refinement, this difference is interpreted at the pipeline level. Together with the local factorial experiment, the results support candidate-source complementarity as the primary mechanism for global TM improvement, while geometry refinement provides a separate benefit in local trace quality.

Keywords: RNA three-dimensional structure, template-based modeling, DRfold2, TM-score, C1′-lDDT, temporal safety, geometry refinement.

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Chapter 1

Introduction

1.1 Background and motivation

RNA regulates gene expression, catalyzes reactions, recognizes molecular partners, and forms ribonucleoprotein complexes. These functions depend not only on the order of A, C, G, and U, but also on the base pairs, motifs, and three-dimensional arrangements formed by the chain [2, 3, 4]. Secondary structure describes base pairing and local motifs, whereas tertiary structure captures how these elements interact in space [3, 4]. An RNA 3D predictor must consequently recover both the global fold and its local geometry.

X-ray crystallography, nuclear magnetic resonance, and cryogenic electron microscopy are major routes to experimental RNA structures. Each technique has a distinct domain of applicability, and cryo-EM has expanded access to large RNA-containing complexes; nevertheless, sample preparation, data acquisition, and model reconstruction remain resource-intensive [1, 5]. The PDB provides the global archive for experimentally determined macromolecular structures, but RNA remains less represented than protein and many RNA families lack close structural templates [6, 1]. This gap motivates computational prediction from sequence.

Current RNA 3D methods draw on physical principles, knowledge extracted from observed structures, and learned patterns. Template-based modeling can be highly accurate when a suitable structural relative is recognized, whereas deep models can produce candidates when direct template retrieval is weak [11, 16, 17, 18]. Blind RNA-Puzzles and CASP assessments show that rankings vary across targets, lengths, fold novelty, and permitted information [7, 9, 10]. A bank of candidates from different sources is therefore attractive when the evaluation retains the best of several predictions.

The Stanford RNA 3D Folding competition provides a particularly relevant setting. More than 1,700 teams predicted 43 previously unreleased structures, and both leading systems used TBM. Post-competition analysis further showed that combining TBM and deep learning could outperform individual components on the same test set [12]. This observation does not make every RNA a template-rich problem. Instead, it motivates two questions: how to retrieve structures without future information, and how to cover targets for which templates are inadequate.

1.2 Research problem

A TBM pipeline can receive an artificially high score if its library contains a structure released after the prediction cutoff. This temporal leakage uses information that is public at rerun time but unavailable in the original blind setting. RNA-Puzzles and CASP explicitly separate prediction from unreleased structures, making the data cutoff part of the scientific protocol [7, 9, 12]. This thesis requires every template to predate its target and excludes the target's own reference PDB entries.

Even under a valid cutoff, a single retrieval system may miss useful templates. MMseqs2 supports fast, sensitive sequence search at database scale [13], but structural relatives do not always present a sufficiently strong homolog for the fast retrieval path. We therefore complement MMseqs2 with a search heuristic based on global alignment, local alignment, sequence composition, and 3-mer overlap. Its purpose is to increase candidate recall rather than define an optimal biological score.

TBM cannot provide a strong model for every target. DRfold2 is therefore introduced as an independent source rather than a replacement for templates. DRfold2 extends the line of work that combines end-to-end learning with learned geometric potentials for RNA 3D prediction [16, 18]. Retaining three TBM and two DRfold2 models preserves template strength on easier targets while reserving two slots for folds that the deep model may recover better.

Candidates from both branches may still contain irregular backbone distances, clashes, or local distortions. Previous nucleic-acid refinement tools show that better geometric quality does not guarantee systematic improvement in native RMSD [20, 21]. The refinement objective in this thesis is therefore narrow: improve local C1′ geometry while keeping TM-score within a predeclared preservation margin. Refinement is not presented as a second global-fold predictor.

1.3 Research questions

The thesis addresses three questions in the order in which information enters the pipeline:

RQ1. What does composite search add to MMseqs2 in a time-safe TBM pipeline, and which ranking components produce measurable changes?

RQ2. When candidate count is controlled, does DRfold2 complement the targets missed by TBM?

RQ3. Does geometry refinement improve C1′-lDDT and sliding-window RMSD while preserving global TM-score?

These questions form a testable progression. RQ1 explains candidate availability, RQ2 tests source diversity, and RQ3 identifies the structural scale affected by postprocessing. The leaderboard is used as an external assessment of the frozen pipeline, not as a substitute for local ablation.

1.4 Contributions

The first contribution is an auditable time-safe TBM branch that unifies fast and composite retrieval, enforces release-date and self-PDB filters, realigns candidates, ranks them by identity and coverage, and transfers C1′ coordinates. Component studies evaluate each decision instead of inferring its value from a single final score.

The second contribution is quantitative evidence of complementarity between TBM and DRfold2. The study reports each source, their union, and fixed-size candidate banks. Fixed-N comparisons partially separate genuine source effects from the mechanical benefit of adding more prediction slots.

The third contribution is conservative geometry refinement evaluated with a held-out protocol. Priors are estimated on 60 RNAs, the configuration is selected on 20, and the method is frozen before opening 20 newer RNAs. Raw candidates, a simple smoothing baseline, the full objective, cumulative ablation, and leave-one-out ablation are all reported.

Finally, the thesis provides a two-level explanation of leaderboard performance. The 0.61390 private score establishes external pipeline value, while local complementarity and refinement experiments identify component-level mechanisms. Keeping these evidence levels separate prevents the refinement stage from receiving unsupported credit for the entire score increase.

1.5 Scope and organization

The system predicts one C1′ coordinate per nucleotide and returns five structures per target. Confirmatory local analysis covers single-chain RNAs up to 96 nucleotides; Kaggle provides an external check on a different distribution that includes longer targets. Full-atom chemistry, ions, ligands, chemical modifications, and RNA-protein interfaces are outside the evaluated representation, although each can be important to RNA structure and function [4, 1].

Chapter 2 reviews the scientific and computational background. Chapter 3 describes the data and leakage controls. Chapter 4 presents the pipeline, and Chapter 5 defines the experiments. Results are reported in Chapter 6, followed by discussion in Chapter 7 and conclusions in Chapter 8.

Chapter 2

Background and Related Work

2.1 RNA structural organization

RNA is a polymer with a sugar-phosphate backbone and four primary bases. Watson-Crick interactions, base stacking, noncanonical pairs, ions, and tertiary contacts contribute to structural stability [3, 4]. Interactions that are distant in sequence can become adjacent in space, allowing a local motif error to alter the molecular arrangement. Local and global quality must therefore be measured with different metrics.

This thesis uses the C1′ trace as a common representation. For a sequence of length L, a candidate is

$$X = [\mathbf{x}_1, \ldots, \mathbf{x}_L]^{\mathrm{T}} \in \mathbb{R}^{L\times 3}, \tag{2.1}$$

where $\mathbf{x}_i$ is the C1′ coordinate of nucleotide i. This coarse-grained form matches the competition output [12, 27]. It does not encode base orientation or full chemistry, so conclusions are limited to fold and C1′-trace quality.

Rigid rotation and translation do not alter internal structure. Kabsch superposition finds the rotation that minimizes squared deviations for known correspondence [25]. lDDT avoids global superposition, whereas TM-score searches for a structural alignment and uses length-dependent normalization [22, 24, 23]. The metrics consequently respond to distinct scales of structural error.

2.2 Template-based modeling

TBM assumes that a known structure can provide a scaffold for a related query. A typical workflow retrieves templates, aligns query and template, transfers corresponding coordinates, builds unsupported regions, and refines the model [11, 12]. Accuracy depends on both recognizing a suitable structure and aligning residues correctly. High identity over a short fragment may be less useful than weaker similarity over nearly the complete target.

Needleman-Wunsch solves global sequence alignment, whereas Smith-Waterman identifies the best local alignment [14, 15]. The two algorithms expose different signals: global alignment measures whole-chain agreement and local alignment recognizes conserved regions. The composite search combines both with composition and 3-mer features to broaden retrieval.

MMseqs2 enables sensitive search over large sequence collections [13]. Retrieval speed, however, is not equivalent to structural coverage. The experiments therefore report availability, conditional quality when a hit exists, and whole-cohort means separately. This prevents the 12 targets without MMseqs2 hits from silently disappearing from the evaluation.

2.3 Deep RNA structure prediction

Recent deep models learn sequence representations, geometric distributions, or coordinates for RNA. ARES applies geometric deep learning to structure scoring, DRfold combines end-to-end prediction with learned potentials, RhoFold+ uses an RNA language model, and DRfold2 develops a more efficient predictive pipeline [19, 16, 17, 18]. Although these approaches reduce direct dependence on a close template, their accuracy remains target and benchmark dependent [10, 18].

DRfold2 is not framed as a universal replacement for TBM in this thesis. The sources have distinct failure modes: TBM depends on retrieval and alignment, whereas a pretrained model depends on its training distribution and learned evolutionary signals. A common candidate bank exploits this difference, and the held-out experiment tests whether the difference is useful in practice.

2.4 Structure refinement

Refinement searches the neighborhood of a candidate for structures that better satisfy geometric or energetic constraints. QRNAS optimizes nucleic-acid models with an extended energy function, and Arena reconstructs full-atom RNA from coarse-grained predictions [20, 21]. This literature distinguishes stereochemical regularization from movement toward the native structure. Better bonds and fewer clashes do not guarantee a better global fold.

The refinement in this thesis follows the same distinction. A source restraint preserves the initial candidate; distance, angle, and torsion priors regularize the trace; and clash, kink, and radius-of-gyration terms target specific artifacts. Effectiveness is measured primarily with TM-score, C1′-lDDT, and sliding-window RMSD rather than the optimized loss itself [22, 24, 23].

2.5 Structural evaluation

TM-score was introduced as a length-normalized measure of structural similarity [22], and US-align extends structural alignment to nucleic acids and macromolecular complexes [23]. The competition scores each target using the best combination of five predictions and available native conformations, then averages over targets [12]. In this thesis, best-of-five is an evaluation oracle only; inference never uses native structures to select a candidate.

lDDT evaluates local distance preservation without global superposition [24]. We compute it on C1′ atoms and denote it C1′-lDDT. Sliding-window RMSD superposes each contiguous window by Kabsch and then measures coordinate deviation; windows of 9 and 15 nucleotides respond more strongly to local distortions than a 31-residue window or whole-structure RMSD [25]. Together, these metrics separate local trace correction from global fold change.

2.6 Research gap

Benchmarking studies compare RNA 3D predictors, and the competition analysis shows that TBM remains competitive as structural databases grow [10, 12]. A leaderboard score, however, does not reveal which component produced the result, and a component study without a temporal cutoff can favor template search. The first gap is therefore a white-box study of TBM under a realistic release-date policy.

The second gap concerns source complementarity under controlled candidate count. A five-model union can beat a three-model bank merely because it has more opportunities. This thesis reports both union and fixed-N comparisons. The final gap is confirmatory evaluation of geometry refinement against raw candidates, a simple baseline, independent metrics, and ablations. These gaps map directly to the three research questions.

Chapter 3

Data and Evaluation Controls

3.1 Data sources and splits

The template library is built from public PDB structures. The PDB is the single global archive for macromolecular structural data and supplies chain, sequence, coordinate, and release metadata [6]. The frozen project snapshot contains 8,670 structure files, 23,869 RNA or RNA-DNA chains, and approximately 10.86 million residues. About 99.9% of residues contain a C1′ coordinate. The MMseqs2 index has roughly 20,844 valid chains, while the deduplicated composite library has 7,155 unique sequences. These counts describe the project artifact rather than the current PDB.

The initial development set contains 12 CASP15 targets of 30 to 720 nucleotides. CASP is a blind assessment in which target structures are unreleased during prediction, and its RNA targets have an official independent assessment [9]. Because this cohort is small and intentionally difficult, it is used for development, leakage illustration, and deployment checks rather than as the only confirmatory evidence.

The principal study uses 100 RNAs released between January 2024 and March 2025. Every target has three blindly generated TBM candidates and two DRfold2 candidates. The data are split temporally into 60 train, 20 calibration, and 20 validation RNAs as shown in Table 3.1. An audit found no sequence group or family group crossing a split.

Table 3.1: Data split for geometry refinement and complementarity analysis.

Split	RNAs	Release-date range	Role
Train	60	2024-01-10 to 2024-12-04	Estimate geometric priors
Calibration	20	2024-12-11 to 2025-01-08	Select and freeze configuration
Validation	20	2025-01-15 to 2025-03-26	One confirmatory evaluation

The dominant feature of Table 3.1 is role separation. Train creates statistics, calibration determines hyperparameters, and validation measures the frozen method. No validation result is used to revise the configuration. This protocol cannot remove every distribution bias, but it prevents the same target from selecting and confirming the method.

Kaggle provides external evaluation with sequences and native structures hidden during notebook execution. The competition requires five predictions per target and scores the best using US-align on C1′ atoms [12, 27]. We retain the public and private scores, execution manifest, and final CSV hash. Because hidden native structures are unavailable, Kaggle supplies a pipeline-level scalar rather than per-target ablation.

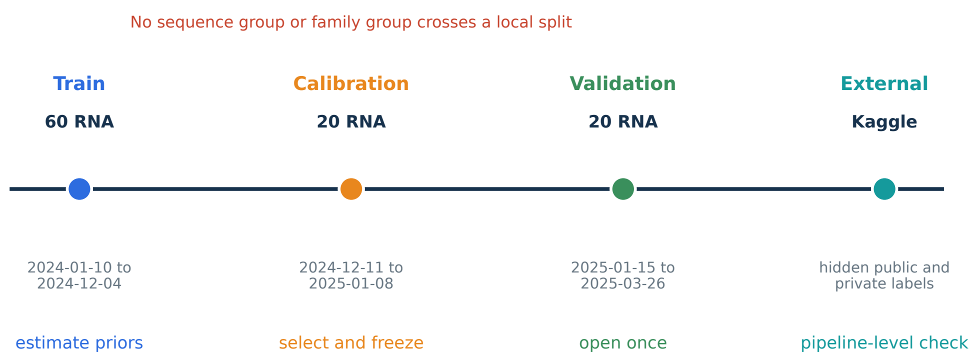

Figure 3.1: Two-level evidence protocol. Local splits are separated by time and family; Kaggle supplies an external hidden-set check of the complete pipeline.

3.2 Temporal and self-template exclusion

For target q with cutoff t_q, template s is valid only if

$$\text{release}(s) < t_q \quad \text{and} \quad \text{PDB}(s) \notin \mathcal{E}_q, \tag{3.1}$$

where $\mathcal{E}_q$ contains the target's excluded PDB entries. A strict inequality avoids ambiguity for same-day releases. Missing cutoffs or release dates should fail closed rather than default to an admissible template.

Figure 3.2 illustrates the consequence of violating this rule on the early CASP15 pipeline. Mean best-of-five TM increased from 0.1612 with the time-safe filter to 0.6388 without it and 0.9566 under an oracle leak. Unsafe variants are not candidate methods; they quantify how an invalid data snapshot can inflate evaluation. The blind principles of CASP and RNA-Puzzles motivate this audit [7, 9].

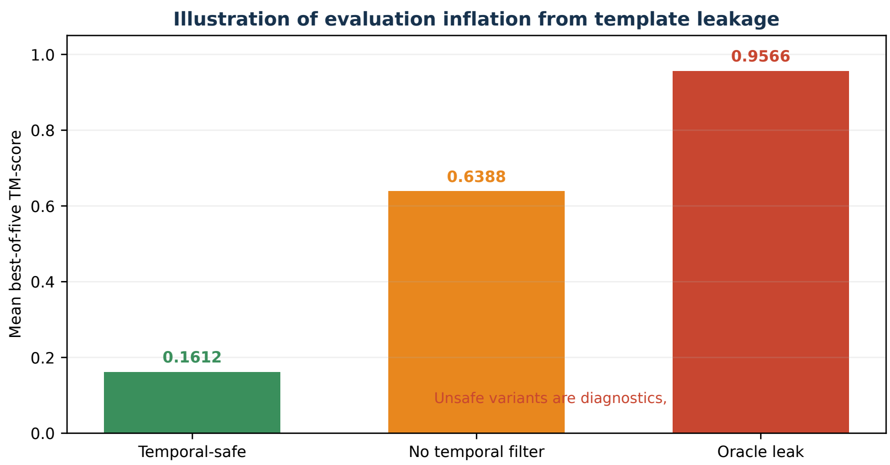

Figure 3.2: Evaluation inflation from template leakage on the development CASP15 cohort. The unsafe variants are diagnostics only.

3.3 Label-use policy

Native labels are read only by local evaluation scripts after candidate generation. Within the complementarity analysis, an "oracle" is a maximum used to measure the available ceiling of a candidate bank; it is never an inference-time selector. Geometry refinement receives only source coordinates, source confidence, and aggregate priors. For each candidate, the reference conformation used for local metrics is fixed from the raw candidate so that refinement cannot benefit by switching to an easier native.

The final Kaggle artifact contains no validation labels, hidden labels, or prebuilt submission. Its manifest records `native_labels_used: false`; the output has 2,515 rows, 18 columns, correct identifiers, and finite coordinates. The final SHA-256 is

```
54fe6e6598a1c2924ef7cbf03482f7ecf3c2e52e8726a1609569f777b1ac8c9a
```

These checks establish deployment integrity, while Equation 3.1 governs template validity.

3.4 Statistical unit

The RNA target is the independent unit for every confidence interval. Candidates from one target share a sequence, native structure, and generation process, and are not treated as independent observations. For target effects d_i, we report

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i \tag{3.2}$$

and a percentile interval from 10,000 bootstrap samples of targets with replacement. Bootstrap resampling provides a nonparametric estimate of uncertainty from the observed sample [26]. The random seed is fixed at 2025.

Effects are oriented so that positive values indicate improvement. TM-score and lDDT use new minus baseline, whereas RMSD, clash, and deviation use baseline minus new. Tables also report medians and improve, tie, and regress counts when available, preventing a mean effect from hiding systematic failures.

Chapter 4

Proposed Pipeline

4.1 Overview

The pipeline accepts an RNA sequence and temporal cutoff, then generates candidates from two independent branches. TBM returns three ranked structures, and DRfold2 returns two structures by internal confidence. When DRfold2 exceeds its resource limit, TBM fills the unused positions. Geometry refinement processes each candidate independently for 300 steps, and the writer emits exactly five C1′ traces.

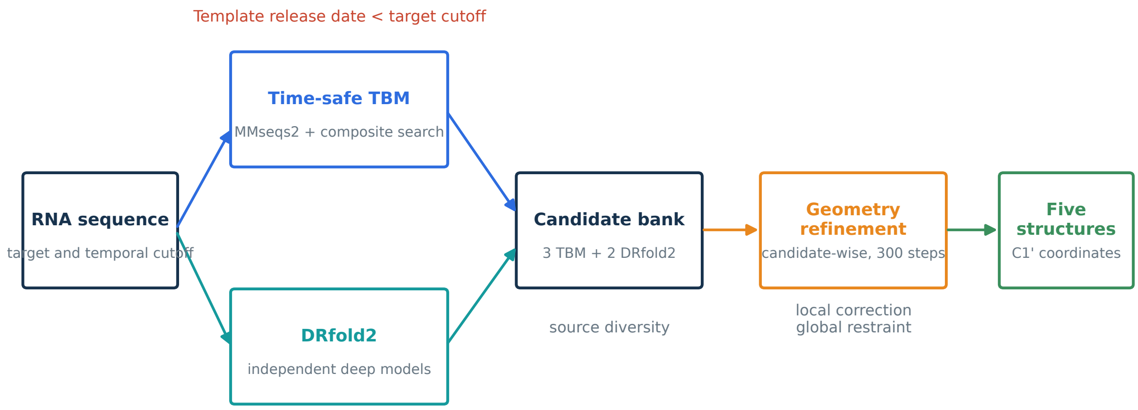

Figure 4.1: Final pipeline. Source diversity is created before candidate-wise refinement, with no native-guided selection or coordinate fusion.

The design follows two principles. First, diversity is introduced at the source level through three TBM and two DRfold2 candidates. Second, refinement remains close to each source to avoid disrupting an already correct fold. The two principles address distinct failure modes and are evaluated separately.

4.2 Time-safe TBM

4.2.1 Dual-path retrieval

The first path runs MMseqs2 over the template-sequence index. MMseqs2 uses accelerated filters and sensitive alignment to search large collections [13]. Raw hits are normalized to chain keys, identity, alignment coverage, and release dates. Hits that lack C1′ coordinates or violate temporal validity are removed before ranking.

The second path scans the deduplicated library with

$$S_{\text{comp}}(q,s) = 0.4G(q,s) + 0.3L(q,s) + 0.2F(q,s) + 0.1K_3(q,s), \tag{4.1}$$

where G is global-alignment similarity, L is local-alignment similarity, F measures sequence-composition agreement, and K_3 is 3-mer overlap. The alignment terms use the Needleman-Wunsch and Smith-Waterman formulations [14, 15]. The weights are frozen heuristics, and their sensitivity is tested by ablation rather than assigned a probabilistic interpretation.

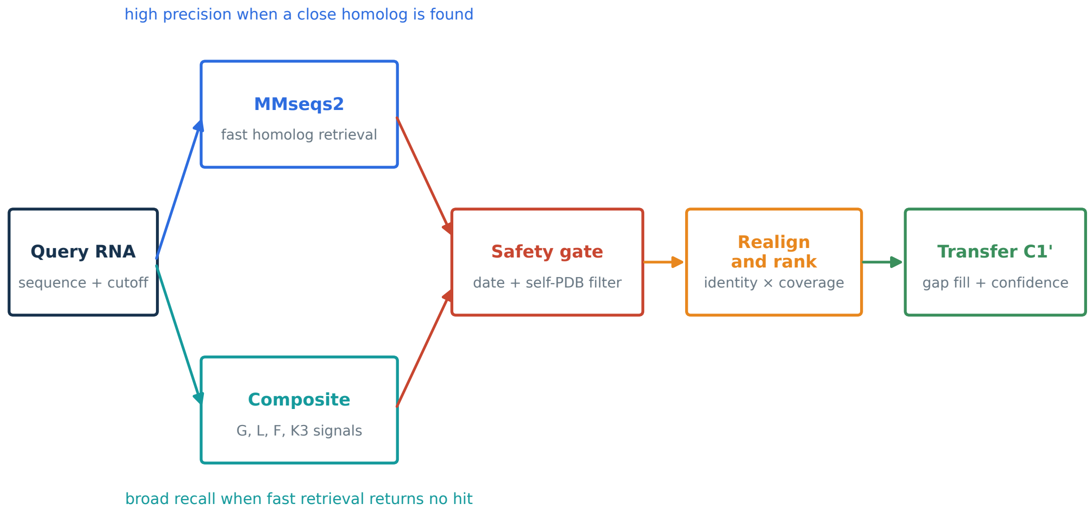

Figure 4.2: TBM flow. MMseqs2 prioritizes clear homologs, composite search broadens recall, and the safety gate precedes realignment and coordinate transfer.

The two lists are merged by chain key before temporal and self-PDB filtering. The design in Figure 4.2 keeps the roles separate: MMseqs2 is not considered inaccurate when no hit is returned, and composite search is not considered accurate merely because it always returns a candidate. Availability and conditional quality are reported separately.

4.2.2 Realignment and ranking

Every valid hit is globally realigned to establish one correspondence for transfer. Let I denote identity on aligned positions, C_q the query fraction with supported coordinates, and C_s template completeness. The rank score is

$$S_{\text{rank}} = I\, C_q\, C_s. \tag{4.2}$$

Identity times coverage defines the primary order, while completeness breaks cases with missing coordinates. Distinct PDB sources are preferred where possible, and ablation tests whether this rule changes oracle TM on the cohort.

The rank score is not a calibrated probability of correctness. It orders candidates without native access. Multiplication by C_q is especially useful when a high-identity local hit covers only a small part of the query, as shown by the reranking experiment in Chapter 6.

4.2.3 Coordinate transfer and gap completion

For aligned residues, the template C1′ coordinate is copied to the corresponding query position. Unsupported residues receive low confidence. Internal gaps are interpolated along a curve constrained by neighboring anchors, while terminal regions are extrapolated from nearby backbone steps. A deterministic de novo trace guarantees finite output when support is insufficient.

Curved filling makes the candidate valid before refinement; it does not reconstruct a new motif from native information. The experiment therefore bins gaps by length and measures SW-RMSD9 only at unsupported residues. Evidence of improvement is confined to gaps of 9 to 20 nucleotides.

4.3 DRfold2 branch

DRfold2 receives the RNA sequence and predicts structure using a pretrained deep model. Its publication describes an efficient RNA 3D prediction pipeline developed from DRfold [16, 18]. This thesis preserves the pretrained inference path and does not fine-tune it on the 100 local RNAs. The two candidates with highest model confidence are converted to the common $L \times 3$ C1′ representation.

The DRfold2 models are generated independently of TBM hits. Independence here means that the final branch does not use native quality or TBM scores to alter DRfold2 coordinates; it does not claim that the pretrained model has no relationship to PDB. Family-disjoint local splits protect the project cohort from obvious overlap, while the model-training cutoff remains an external-validity consideration.

The frozen implementation limits DRfold2 to 600 nucleotides. During public execution it ran on 11 of 12 targets and skipped one 720-nucleotide sequence, which retained five TBM candidates. The fallback rule was fixed before hidden evaluation.

4.4 Geometry refinement

4.4.1 Objective

Given source candidate $X^{(0)}$, the refinement finds a nearby X that reduces

$$\mathcal{L}(X) = \lambda_s \mathcal{L}_{\text{source}} + \lambda_b \mathcal{L}_{\text{backbone}} + \lambda_a \mathcal{L}_{\text{angle}} + \lambda_t \mathcal{L}_{\text{torsion}} + \lambda_c \mathcal{L}_{\text{clash}} + \lambda_k \mathcal{L}_{\text{kink}} + \lambda_r \mathcal{L}_{R_g}. \tag{4.3}$$

Each term addresses a specific failure mode, and the source term limits global drift. Figure 4.3 is a conceptual illustration rather than an experimental target.

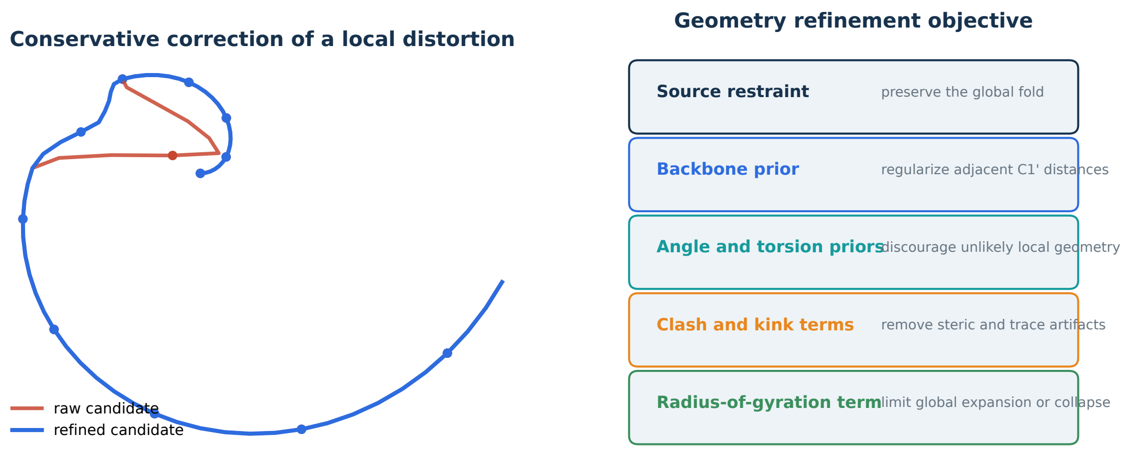

Figure 4.3: Geometry refinement mechanism. Local distortion is regularized while a source restraint preserves the overall arrangement.

4.4.2 Geometric priors

Adjacent distance is $d_i = \|\mathbf{x}_{i+1} - \mathbf{x}_i\|_2$. On the 60 train RNAs, its mean is 6.09 Å, standard deviation is 1.11 Å, and median is 5.66 Å. The non-adjacent C1′ clash threshold is 4.18 Å, and the radius-of-gyration prior is

$$\widehat{R}_g(L) = 5.18 L^{0.346}. \tag{4.4}$$

These values are empirical priors of the frozen project snapshot, not universal physical constants. They are estimated only on train and held fixed thereafter.

The backbone term uses a context-dependent negative log likelihood of adjacent distance. Angle and torsion terms operate on consecutive triples and quadruples. Kink and clash terms penalize sharp bends and close nonadjacent points, while R_g limits global collapse or expansion. Context-dependent distributions allow more flexibility than one fixed target distance.

The confidence-weighted source restraint is

$$\mathcal{L}_{\text{source}} = \frac{1}{L}\sum_{i=1}^{L} w_i \|\mathbf{x}_i - \mathbf{x}_i^{(0)}\|_2^2. \tag{4.5}$$

Weights are larger for template-supported or high-confidence residues, allowing gaps and weak regions to move more than reliable scaffolds. Every candidate is optimized for 300 Adam steps using the configuration frozen on calibration.

4.4.3 Baseline and output

The simple baseline contains only source restraint and backbone smoothing. It tests whether the additional objective terms improve beyond a conservative local smoother. The full method is compared with both raw and simple candidates; improvement over raw alone does not establish the necessity of every term.

After refinement, each candidate is checked for shape, finite values, and residue order. The writer emits five coordinate triples per nucleotide in exact sample-ID order. The final pipeline contains no native-guided router, source selection, or coordinate fusion.

4.5 Algorithm summary

1. Read sequence q, temporal cutoff t_q, and excluded PDB identifiers.
2. Retrieve MMseqs2 and composite hits, then merge by chain key.
3. Remove hits that violate Equation 3.1.
4. Realign, rank, and build three TBM candidates.
5. Run DRfold2 when $L \leq 600$ and retain two candidates by model confidence.

6. Fill unavailable DRfold2 positions with TBM, maintaining five candidates.
7. Refine each candidate independently for 300 steps.
8. Validate identifiers, dimensions, uniqueness, and finite coordinates, then write CSV.

Native labels do not appear in this inference path. All pre-output decisions use sequence, template metadata, pretrained confidence, and aggregate priors. Oracle scores are computed only after generation for analysis.

Chapter 5

Experimental Design

5.1 Testing strategy

Each research question is assigned a distinct experiment and maximum claim. RQ1 uses the 20 calibration RNAs to study retrieval sources, composite terms, reranking, and gap completion. RQ2 uses the 20 validation RNAs to compare candidate banks before refinement. RQ3 follows the full 60/20/20 protocol and opens validation only after configuration freeze. This separation prevents exploratory TBM analyses from serving as confirmatory refinement evidence.

Table 5.1: Mapping from research questions to experiments and claims.

RQ	Primary comparison	Supported claim
RQ1	MMseqs2, composite, union; component and reranking ablations	Composite expands availability; ranking effects are measured by component
RQ2	3T, 2D, union; fixed-N allocation	TBM and DRfold2 complement one another by target
RQ3	Raw, source+backbone, full refinement; cumulative and leave-one-out analyses	Refinement improves local trace within a TM preservation margin

Table 5.1 constrains interpretation before results are examined. RQ3 does not hypothesize an increase in TM-score. The validation gate passes when a primary local metric improves and mean TM change remains within ± 0.005, a margin frozen on calibration.

5.2 TBM ablations

The search-source study compares MMseqs2-only, composite-only, and their union. MMseqs2 TM is conditional on returning a hit, and availability is reported alongside it. Composite and union means use all 20 targets. This avoids comparing eight easier targets with a full 20-target cohort as if the samples were equivalent.

The component study compares G, $G + L$, $G + L + F$, an equal four-term score, the full weighted score, and $\pm 25\%$ perturbations to individual weights. Useful recall at TM 0.45 is measured within a common candidate pool as an internal diagnostic. An unsafe-date variant quantifies leakage and is never considered a valid method.

The reranking study begins with identity, then adds query coverage, template completeness, and distinct-PDB selection. Gap completion compares linear and curved interpolation in length bins 1-3, 4-8, 9-20, and above 20 nucleotides. Gaps within a target are averaged before target-level bootstrap.

5.3 Complementarity analysis

Source candidates are evaluated before refinement. For target i, let T_i contain the three TBM scores and D_i the two DRfold2 scores. The source and union ceilings are

$$O_T(i) = \max T_i, \quad O_D(i) = \max D_i, \quad O_U(i) = \max(T_i \cup D_i). \tag{5.1}$$

The union gain $O_U - O_T$ measures the available value of two additional DRfold2 models. Because the union has more candidates, it combines source diversity and count effects.

Fixed-N comparisons control this issue. At $N = 2$, $2T$ is compared with $1T + 1D$; at $N = 3$, $3T$ is compared with $2T + 1D$. Candidates are ordered by blind source confidence. The cache contains only three TBM and two DRfold2 models per target, so a complete five-slot sweep from $5T$ to $5D$ is not identifiable without duplicating candidates.

5.4 Geometry-refinement analysis

Train estimates priors only. On calibration, cumulative ablation adds source, backbone, angle, torsion, clash, kink, and R_g terms, while leave-one-out analysis removes each term from the full objective. The selected configuration satisfies local improvement and TM preservation, then is serialized before validation.

Validation compares raw candidates, simple source-plus-backbone smoothing, and the full objective. Primary local metrics are C1′-lDDT, SW-RMSD9, and SW-RMSD15. Best-of-five TM is the global safeguard, with whole-structure C1 RMSD and SW-RMSD31 as secondary measures. Backbone deviation, clash, sharp kink, angle NLL, torsion NLL, and R_g error are mechanism diagnostics because they lie in or near the objective.

A fixed-$N = 3$ factorial forms four cells: $3T$ raw, $3T$ refined, $2T + 1D$ raw, and $2T + 1D$ refined. It estimates source-bank gain with refinement off, refinement effect within each bank, and the interaction. This design directly tests whether refinement can explain the hybrid TM increase.

5.5 External evaluation and reproducibility

Two frozen submissions are compared: time-safe composite TBM and the final hybrid. The hybrid replaces two TBM positions with DRfold2 when feasible, then applies the frozen refinement configuration. Public and private scores are read from Kaggle, while the execution manifest and output hash provide integrity checks [27].

Repository scripts generate all local tables. Bootstrap seed is 2025, and frozen configurations, prior provenance, and candidate caches accompany the artifacts. Audits require one metadata row per target, no cross-split groups, exactly three TBM and two DRfold2 candidates per target, finite coordinates, and a fixed reference conformation between raw and refined evaluation.

Chapter 6

Results

6.1 RQ1: TBM retrieval and ranking

6.1.1 Retrieval sources

MMseqs2 found a template for 8 of 20 calibration RNAs, an availability of 40%. On those eight targets, top-1 TM was 0.660029 and best-of-five TM was 0.678316. Composite search produced candidates for all 20 targets but reached a lower whole-cohort mean of 0.376060. The union maintained full availability and increased the mean to 0.389705.

Table 6.1: Retrieval-source results on 20 calibration RNAs. MMseqs2 quality is conditional on the eight targets with a hit.

Source	Targets with hits	Availability	Best-of-five TM when available
MMseqs2-only	8/20	0.40	0.678316
Composite-only	20/20	1.00	0.376060
MMseqs2 + composite	20/20	1.00	0.389705

The sources have distinct roles. MMseqs2 candidates are strong when a homolog is recognized, while composite search fills the remaining 12 targets at lower average precision. The union gained 0.013645 over composite-only, although its interval $[-0.004417, 0.042781]$ included zero. The clearest benefit is therefore availability, not uniform superiority of composite hits.

6.1.2 Composite components

The full weighted score reached 0.376060, whereas global alignment alone reached 0.381177. The G-only advantage was 0.005117 with confidence interval $[-0.002178, 0.015004]$. Perturbing individual weights by $\pm 25\%$ produced scores from 0.375436 to 0.376500. The heuristic is locally stable but does not clearly outperform global alignment on this cohort.

The $G + L + F$ variant nevertheless had useful recall 0.676304 compared with 0.635488 for G-only within the common hit pool. This pattern is consistent with a recall heuristic: additional signals introduce distinct candidates even when mean oracle TM does not rise clearly. Useful recall remains an internal diagnostic rather than a replacement for TM.

Removing the temporal filter increased mean TM from 0.376060 to 0.403022, a change of 0.026962. The interval $[-0.000343, 0.071258]$ was wide and 15 targets were unchanged. This is not evidence for an unsafe method; it shows that a few leaked templates can shift the cohort mean.

6.1.3 Reranking and gap completion

Adding query coverage to identity increased top-five TM from 0.371900 to 0.387558, a gain of 0.015659 with interval $[-0.003731, 0.045178]$. Completeness and distinct-PDB selection did not change oracle TM on this cohort, although the latter increased the mean number of distinct PDB sources. Coverage is therefore the dominant ranking signal in Table 6.2, while later rules primarily support robustness.

Table 6.2: TBM reranking ablation.

Ranking	Top-1 TM	Top-3 TM	Top-5 TM
Identity only	0.334450	0.363854	0.371900
Identity $\times$ coverage	0.363720	0.378412	0.387558
$+$template completeness	0.363720	0.378412	0.387558
$+$distinct PDB	0.363720	0.378412	0.387558

Curved gap filling showed clear improvement only for gaps of 9 to 20 nucleotides. It reduced SW-RMSD9 by 0.036778 Å with interval $[0.009289, 0.067605]$. Bins 1-3 and 4-8 showed no clear difference from linear filling, and gaps above 20 were unchanged. The curve has enough span to avoid a straight bridge in the middle bin but insufficient information to reconstruct very long missing motifs.

6.2 RQ2: Source complementarity

Before refinement, three TBM candidates reached a mean oracle of 0.565183, and two DRfold2 candidates reached 0.502627. Their five-candidate union reached 0.588304, a 0.023121 gain over TBM with interval $[0.006444, 0.047519]$. DRfold2 supplied the better source for 8 targets, whereas TBM did so for 12.

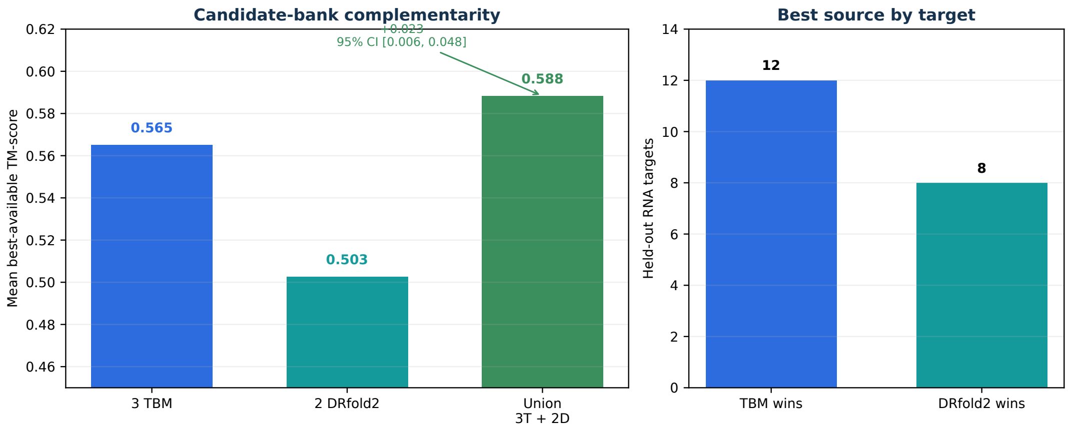

Figure 6.1: Candidate-bank complementarity on 20 validation RNAs. The union interval is obtained by target-level bootstrap.

The central result is not higher mean accuracy for DRfold2. Its source mean was 0.062556 below TBM. Its value arose from the eight targets on which source ordering reversed. Because oracle union cannot decrease when candidates are added, fixed-N comparisons are required to separate source from count effects.

At two slots, $1T + 1D$ exceeded $2T$ by 0.030546 with interval $[0.002949, 0.068185]$. At three slots, $2T + 1D$ exceeded $3T$ by 0.016012, although the interval $[-0.002072, 0.041466]$ included zero and ten targets tied. The declining marginal effect suggests that DRfold2 is most useful when it replaces a redundant template; three diverse TBM candidates leave less room for improvement.

Table 6.3: Candidate allocation with geometry refinement disabled.

Comparison	Baseline mean	Hybrid mean	Mean gain and 95% CI
$2T$ versus $1T + 1D$	0.549040	0.579586	$+0.030546$ $[0.002949, 0.068185]$
$3T$ versus $2T + 1D$	0.565183	0.581196	$+0.016012$ $[-0.002072, 0.041466]$
$3T$ versus $3T + 2D$	0.565183	0.588304	$+0.023121$ $[0.006444, 0.047519]$

Table 6.3 distinguishes fixed-size and production-bank effects. In both the two-slot and three-slot comparisons, replacing one TBM candidate with DRfold2 increased the mean, but only the two-slot confidence interval lay entirely above zero.

6.3 RQ3: Geometry refinement

6.3.1 Confirmatory results

Full refinement increased C1′-lDDT from 0.659836 to 0.666496, a gain of 0.006660 with interval $[0.000933, 0.013093]$; 12 of 20 targets improved. SW-RMSD9 decreased from 2.651395 to 2.611010 Å, an improvement of 0.040386 Å with interval $[0.029354, 0.052090]$ and 19 targets improving. SW-RMSD15 improved by 0.017257 Å with interval $[0.000617, 0.031336]$.

Global TM changed from 0.588304 to 0.587197, or -0.001107 with interval $[-0.002809, 0.000418]$. The effect remained inside the predeclared 0.005 preservation margin. Whole-structure C1 RMSD and SW-RMSD31 showed no clear change.

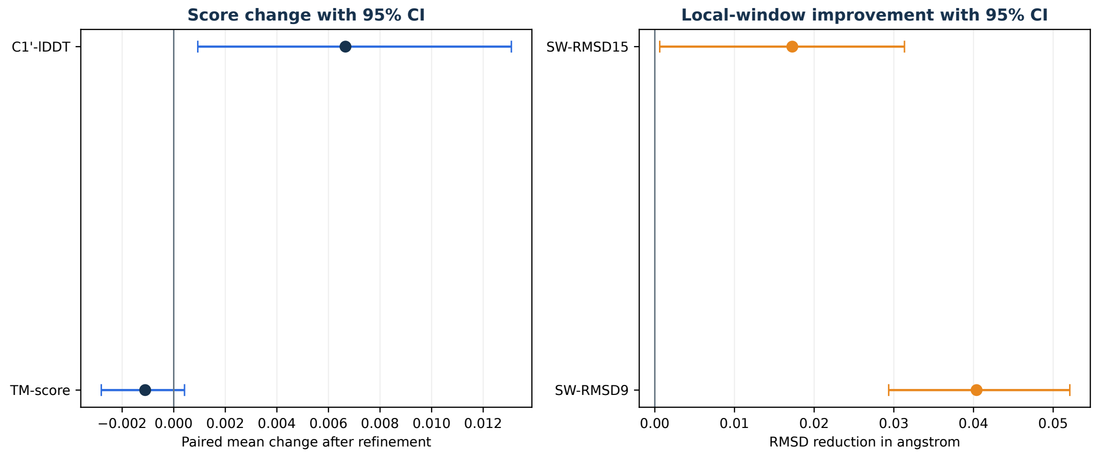

Figure 6.2: Confirmatory geometry-refinement effects. Score changes and RMSD reductions use separate axes and units.

Table 6.4: Raw and full geometry refinement on 20 validation RNAs.

Metric	Raw	Refined	Improvement	95% CI
Best-of-five TM	0.588304	0.587197	-0.001107	$[-0.002809, 0.000418]$
C1′-lDDT	0.659836	0.666496	$+0.006660$	$[0.000933, 0.013093]$
SW-RMSD9 (Å)	2.651395	2.611010	$+0.040386$	$[0.029354, 0.052090]$
SW-RMSD15 (Å)	4.273393	4.256136	$+0.017257$	$[0.000617, 0.031336]$
Backbone deviation (Å)	1.040270	0.785817	$+0.254453$	diagnostic
Clash per residue	0.065121	0.022802	$+0.042319$	diagnostic

The dominant pattern in Table 6.4 is stronger improvement at shorter spatial scales. SW-RMSD9 is the most consistent effect, SW-RMSD15 is smaller, and SW-RMSD31 and global RMSD are inconclusive. This scale gradient matches the design: refinement adjusts neighboring relationships rather than rearranging the fold.

6.3.2 Simple baseline and ablation

Source-plus-backbone smoothing increased lDDT by 0.007581, numerically above the full method's 0.006660. The full-minus-simple difference was -0.000921 with interval $[-0.003118, 0.001327]$, so the full objective did not outperform the simple baseline on lDDT. In contrast, it improved SW-RMSD9 by a further 0.030585 Å with interval $[0.019640, 0.044115]$ and corrected angle, torsion, clash, and kink diagnostics that the simple baseline degraded.

Leave-one-out analysis linked terms to mechanisms. Removing backbone nearly eliminated the lDDT gain; removing angle or torsion degraded the corresponding NLL; removing kink increased sharp bends; and removing clash or R_g worsened clash diagnostics. These results show that the objective behaves as designed, but optimized diagnostics are not treated as independent evidence. The primary claim remains based on lDDT and sliding-window RMSD.

6.3.3 Source-by-refinement factorial

Within the $3T$ bank, refinement changed TM by -0.005852 with interval $[-0.014781, 0.000724]$. Within $2T + 1D$, the effect was -0.002604 with interval $[-0.005473, -0.000278]$. The interaction was $+0.003248$ with an interval containing zero. Table 6.5 therefore shows a positive source-bank effect before refinement and no positive TM effect from refinement in either bank.

Table 6.5: Fixed-$N = 3$ factorial separating source composition and geometry refinement.

Candidate bank	Raw mean TM	Refined mean TM
$3T + 0D$	0.565183	0.559331
$2T + 1D$	0.581196	0.578592

The factorial contradicts a simple explanation in which refinement causes the hybrid global-TM gain. Replacing one of three TBM candidates with DRfold2 already produced a 0.016012 mean increase with refinement disabled. Refinement slightly reduced TM in both banks. Source diversity is therefore the better-supported global mechanism, while refinement acts on local quality.

6.3.4 Subgroups and failure cases

In a post hoc stratification, initially low-quality candidates gained 0.016261 lDDT with interval $[0.008900, 0.024089]$. High-quality candidates changed by -0.001788 with interval $[-0.005775, 0.002420]$. This pattern motivates a future blind quality gate but cannot revise the current method because the groups were examined after validation.

Three failures remain visible. Targets 9E2Z_F and 8K7W_A lost about 0.009 TM, while 9B0S_Et lost about 0.011 lDDT. Conservative refinement does not imply monotonic improvement for every target, and both uncertainty and failure counts are needed beside the mean.

6.4 External Kaggle evaluation

Time-safe composite TBM scored 0.60084 public and 0.60175 private. The final pipeline reached 0.62809 and 0.61390, absolute gains of 0.02725 and 0.01215. The reference TBM-only notebook reported 0.59298 private [28]; the final pipeline was 0.02092 higher.

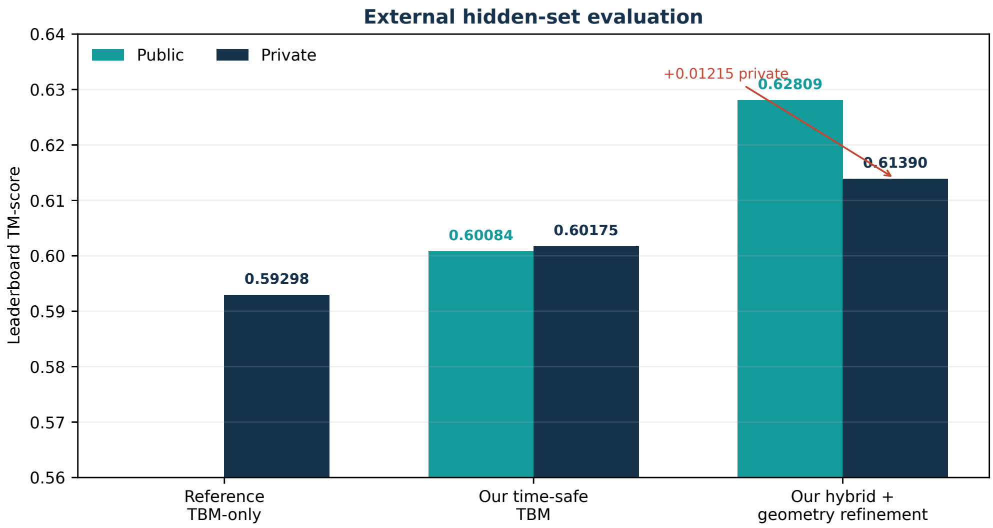

Figure 6.3: Hidden-set evaluation. Only a private score is reported for the reference notebook; both public and private scores are available for the thesis pipelines.

The public execution provides deployment checks. DRfold2 generated two candidates for 11 of 12 targets, the 720-nucleotide target used five TBM candidates, and refinement completed for all 60 structures. Output shape and identifiers were valid, and the run did not read native labels. These observations support execution integrity but cannot identify which hidden targets produced the score increase.

The Kaggle result must be read with the local factorial. The final submission both replaced two TBM slots and applied refinement, making the private change a pipeline-level effect. Local evidence shows that hybrid composition improves global TM before refinement, whereas refinement improves local metrics and slightly reduces TM. Together, the two evidence levels explain the higher score without attributing it wholesale to geometry refinement.

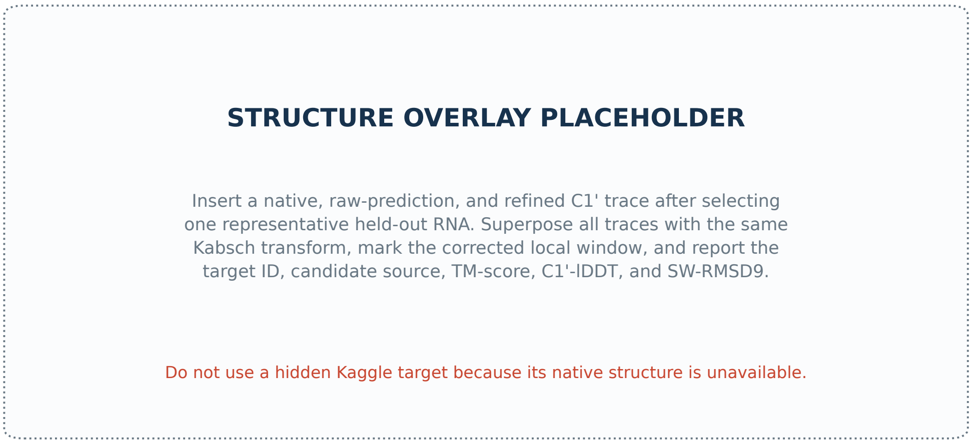

Figure 6.4: Controlled placeholder for a native, raw, and refined structure overlay. The final figure should use a public validation RNA with complete provenance, never a hidden Kaggle target.

Figure 6.4 is intentionally a placeholder rather than a fabricated example. A representative target should be selected by a declared rule, such as SW-RMSD9 change nearest the median. The final caption should record target identifier, candidate source, raw and refined metrics, highlighted window, and generation script.

Chapter 7

Discussion

7.1 Why the hybrid pipeline has value

The gain over the TBM-only reference does not require one source to dominate everywhere. TBM had the higher source mean and won on 12 targets, while DRfold2 won on the remaining eight. Under best-of-five scoring, a distinct source only needs to win a substantial subset of targets to improve the union mean. Complementarity is therefore a target-wise property rather than a comparison of source averages.

The two-slot fixed-N result strengthens this argument by removing candidate-count advantage. Replacing one of two TBM structures with a DRfold2 structure retained a positive confidence interval. The three-slot effect was weaker because three TBM models already contained more diversity and the validation cohort contained only 20 targets. The three-plus-two allocation is consequently a reasonable compromise for the hidden distribution, not a universal optimum.

Competition data provide a second explanation. The analysis reported that 19 of 20 private targets had a prior structure with TM-align above 0.45, a higher template-rich fraction than CASP15 [12]. A large template snapshot and broad retrieval were advantageous in this regime. The same paper cautions that this ordering may not generalize to template-poor RNA [12]; DRfold2 remains an important hedge against that part of the distribution.

7.2 Why refinement does not increase TM-score

The source restraint places the optimum near the starting candidate. Backbone, angle, torsion, and clash terms operate mainly on short-range relationships and contain no signal for changing topology or moving an entire domain. Improvement in lDDT and SW-RMSD9 with nearly unchanged TM is therefore consistent with the intended mechanism.

An incorrect global fold requires long-range contacts, motif information, or a different template. Local optimization around one source cannot invent this evidence. Nucleic-acid refinement studies likewise distinguish improved stereochemistry from systematic native RMSD improvement [20]. Geometry refinement should therefore be interpreted as trace regularization, with source diversity setting the primary global ceiling.

The simple baseline matched the full method on lDDT, identifying backbone smoothing as the principal mechanism for that score. The full objective showed clearer value for SW-RMSD9 and balanced angle, torsion, clash, and kink diagnostics. This narrows the contribution: the full objective is not necessary for every local gain, but it provides a more balanced trace without disrupting the average fold.

7.3 Interpreting the leaderboard

The 0.61390 private score has value because it is measured on hidden structures and exceeds both the project's TBM pipeline and the reference notebook. It establishes that the complete implementation transfers beyond the local cohort. A scalar score, however, does not reveal the per-target counterfactual obtained by disabling one component.

A defensible explanation therefore has three layers. The integrity audit shows that the submission does not use native labels and enforces temporal filtering. The local complementarity analysis identifies a positive global mechanism. The local refinement analysis identifies a separate local mechanism and shows that it does not increase TM. Together, these layers explain performance while preserving the distinction between external validity and component causality.

Distribution shift is also visible. The frozen hybrid rule scored 0.46450 on the 12 CASP15 targets and was worse than the local TBM bank, whereas it improved Kaggle private TM by 0.01215. This reversal shows that effective allocation depends on the target distribution. Reporting both outcomes is more credible than selecting the favorable cohort and argues against a universally optimal mixture.

7.4 Threats to validity

7.4.1 Internal validity

Calibration and validation contain only 20 targets each, leaving wide intervals for several ablations. Target bootstrap handles dependence among candidates but cannot create information beyond the observed RNAs [26]. Multiple subgroup analyses also increase the chance of incidental patterns, so they are labeled post hoc and do not change the current method.

The cache contains only three TBM and two DRfold2 candidates. Fixed-N comparisons up to three positions are identified, but a full five-position allocation sweep is not. Future work should generate at least five independent candidates from each source before opening a new cohort.

7.4.2 External validity

Local validation covers single-chain RNA up to 96 nucleotides. Refinement results do not directly generalize to 720-nucleotide RNA, multichain complexes, or RNA-protein interfaces. Such systems have different interaction patterns and experimental uncertainties [1, 5].

The Kaggle private set is unusually template rich [12]. Its score supports the competition distribution but does not guarantee the same ordering on a template-poor benchmark. A new temporal benchmark released after 2026 would provide a stronger generalization test.

7.4.3 Metric validity

TM-score and C1′-lDDT evaluate a coarse-grained trace. They do not directly measure base orientation, hydrogen bonding, ion coordination, or full-atom stereochemistry, all of which contribute to tertiary interactions and RNA function [4, 2]. A high-scoring trace is therefore not yet a complete model for docking or mechanistic interpretation.

Angle NLL, torsion NLL, clash, and kink are objective diagnostics and favor the full method by construction. They explain mechanism but cannot replace independent metrics. Prioritizing lDDT, sliding-window RMSD, and TM reduces circularity, while full-atom validation remains necessary for biochemical use.

7.5 Future work

The nearest extension is quality-gated refinement. Post hoc results suggest larger lDDT gains for weak candidates than strong ones. A blind gate could skip refinement when source confidence and self-consistency are high, but it must be preregistered on a new calibration set and tested on an unopened cohort.

A second direction is greater within-source diversity. Five independent TBM and five DRfold2 models would support a complete allocation sweep and measure the marginal return of each position. A third direction is full-atom reconstruction with an evaluated tool such as Arena, followed by stereochemical metrics designed for RNA [21, 8].

Finally, temporal metadata should fail closed, and self-PDB exclusion should parse target reference identifiers directly. These changes improve auditability rather than score. In a field with continuously released structural data, reconstructing what was known at prediction time is essential to meaningful evaluation.

Chapter 8

Conclusion

This thesis developed and evaluated an RNA 3D pipeline composed of time-safe TBM, DRfold2, and geometry refinement. Rather than treating the final score as the only evidence, the study separated retrieval, source diversity, and local postprocessing into three questions with distinct ablations and protocols.

For RQ1, MMseqs2 produced strong candidates when a homolog was found but covered only 8 of 20 calibration RNAs. Composite search expanded availability to all 20 targets. The full weighted score did not clearly beat global alignment alone, and query coverage was the most influential reranking component. Composite search is therefore supported as a recall heuristic within time-safe TBM.

For RQ2, TBM had higher average source quality, but TBM and DRfold2 won on different targets. Their union improved mean best-available TM by 0.023121 with a positive confidence interval, and the fixed-$N = 2$ comparison retained a 0.030546 gain. These results directly show that source diversity can improve best-of-five performance without one source universally outperforming the other.

For RQ3, geometry refinement increased C1′-lDDT by 0.006660, reduced SW-RMSD9 by 0.040386 Å, and changed TM by -0.001107. A simple smoother explained much of the lDDT gain, while the full objective improved short-window RMSD and geometric diagnostics. The supported conclusion is local trace correction with average global-fold preservation.

The final pipeline reached 0.62809 public and 0.61390 private, exceeding the project TBM pipeline and the reference TBM-only score. The local factorial identifies candidate complementarity as the better-supported mechanism for global improvement, while geometry refinement provides a separate local benefit. This distinction is the thesis's central scientific value: it reports a competitive system while stating precisely which evidence supports each mechanism and where that evidence ends.

Appendix A

Reproducibility Artifacts

The principal artifacts are:

- `scripts/run_tbm_whitebox_ablation.py` for the TBM component study;
- `scripts/summarize_hybrid_complementarity.py` for source allocation;
- `reports/thesis_notes/tbm_whitebox_ablation.md` for RQ1 tables;
- `reports/thesis_notes/hybrid_complementarity.md` for RQ2 tables;
- `reports/thesis_notes/kaggle_submission_analysis.md` for external scores and manifests;
- `scripts/generate_thesis_v2_figures.py` for all data figures in this draft.

The confirmatory runner, its detailed RQ3 report, and the frozen configuration remain in the repository. The final release should add a manifest with the Git commit, Python environment, template-database checksum, pretrained-model version, and exact command for every table.

Appendix B

Figure Completion Checklist

Figure 6.4 is the only intentional placeholder. To complete it, select one validation target by a fixed criterion; read its public native structure, raw candidate, and refined candidate; apply one correspondence and Kabsch convention; plot the three traces with consistent residue indices; mark a representative local window; and record source paths and metrics in the caption.

All other figures are generated from frozen aggregates by the script listed in Appendix A. Before submission, check fonts, grayscale contrast, text size at 100%, and every plotted number against its source table.

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